

2 Differentiation

Let C be the curve of the equation $y = f(x)$. Suppose that we want to find the gradient of the tangent line to C at the point $P(a, f(a))$. Consider a nearby point $Q(x, f(x))$, where $x \neq a$. See Figure 2.1(a).

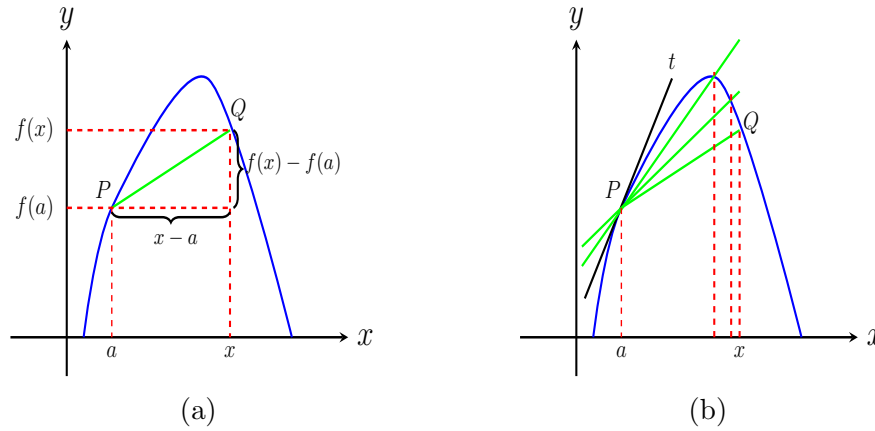


Figure 2.1

The slope of the secant line PQ is

$$m_{PQ} = \frac{\Delta y}{\Delta x} = \frac{f(x) - f(a)}{x - a}. \quad (2.1)$$

From Figure 2.1(b), we observe that x approaches a as Q approaches P along the curve C . The secant line PQ approaches the tangent line t as Q approaches P . Thus, the gradient m_{PQ} of the secant line PQ approaches the gradient m_t of the tangent line t .

Definition 2.1 (Tangent Line). The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with the slope

$$m_t = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

provided that the limit exists.

Now, let $h = x - a$. Then $x = a + h$. It follows that

$$m_{PQ} = \frac{f(a + h) - f(a)}{h}.$$

Besides, we see that $h \rightarrow 0$ as $x \rightarrow a$. Consequently, we infer

$$m_t = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}. \quad (2.2)$$

This is illustrated in Figure 2.2.

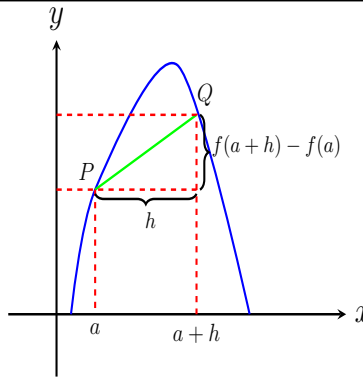


Figure 2.2

Example 2.1. Find the gradient of the tangent line to the graph of $f(x) = x^2$ at the point $P(1, 1)$.

Solution. Let m be the gradient of the tangent line to the graph of $f(x) = x^2$ at the point $P(1, 1)$. Then, we get

$$m = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1^2}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} = \lim_{h \rightarrow 0} (h + 2) = 2.$$

■

Definition 2.2. The derivative of a function f at a number a , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \quad (2.3)$$

if the limit exists.

NB: In Example 2.1, we found $f'(1)$.

Definition 2.3 (The Derivative). The derivative of a function f with respect to x is the function f' defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (2.4)$$

if the limit exists.

The process of finding the derivative of a function is called **differentiation**. There are other notations of the derivative: Let $y = f(x)$. Then, we have that

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x).$$

D and $\frac{d}{dx}$ are called differentiation operators. Additionally, we have

$$f'(a) = \left. \frac{dy}{dx} \right|_{x=a} \text{ or } \left. \frac{dy}{dx} \right]_{x=a}.$$

Definition 2.4 (Left Differentiability). Let f be a function and a a point at which f is defined. Then, f is said to be left differentiable at a if

$$\lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} \quad (2.5)$$

exists. This is denoted by

$$f'(a^-) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}. \quad (2.6)$$

Definition 2.5 (Right Differentiability). Let f be a function and a a point at which f is defined. Then, f is said to be right differentiable at a if

$$\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} \quad (2.7)$$

exists. We write this as

$$f'(a^+) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}. \quad (2.8)$$

Theorem 2.1. Let f be a function and a a point at which f is defined. Then f is differentiable at a if and only if f is right and left differentiable at a and

$$\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}. \quad (2.9)$$

Example 2.2. Determine whether the function f defined by $f(x) = |x|$ is differentiable at $x = 0$.

Solution. We get

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} -1 = -1$$

and

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} 1 = 1.$$

We find that

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} \neq \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0},$$

which means that f is not differentiable at $x = 0$. ■

Example 2.3. Find the values of the real constants a and b that makes the real-valued function f defined by

$$f(x) = \begin{cases} ax + b, & \text{if } x \leq 3 \\ x^2, & \text{if } x > 3 \end{cases}$$

differentiable at $x = 3$.

Solution. For f to be differentiable at $x = 3$, we must have that

$$\lim_{x \rightarrow 3^-} \frac{f(x) - f(3)}{x - 3} \text{ exists,}$$

$$\lim_{x \rightarrow 3^+} \frac{f(x) - f(3)}{x - 3} \text{ exists}$$

and

$$\lim_{x \rightarrow 3^-} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3^+} \frac{f(x) - f(3)}{x - 3}. \quad (2.10)$$

Now, we note that

$$\begin{aligned} \lim_{x \rightarrow 3^-} \frac{f(x) - f(3)}{x - 3} &= \lim_{x \rightarrow 3^-} \frac{ax + b - (3a + b)}{x - 3} \\ &= \lim_{x \rightarrow 3^-} \frac{ax - 3a}{x - 3} \\ &= \lim_{x \rightarrow 3^-} a \\ &= a. \end{aligned}$$

Furthermore, we infer

$$\lim_{x \rightarrow 3^+} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3^+} \frac{x^2 - (3a + b)}{x - 3}. \quad (2.11)$$

Since f must be right differentiable at $x = 3$, that is, the limit in Equation (2.11) exists, we deduce that

$$3a + b = 9. \quad (2.12)$$

Consequently, we arrive at

$$\lim_{x \rightarrow 3^+} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3^+} x + 3 = 6.$$

In view of Equation (2.10), we obtain $a = 6$. Substituting $a = 6$ in Equation (2.12), we get $b = -9$. ■

Theorem 2.2. If f is differentiable at a , then f is continuous at a .

Proof. Suppose f is differentiable at a . Then, it follows that

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists. For $x \neq a$, we can write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a),$$

which yields

$$\lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \right) \cdot \lim_{x \rightarrow a} (x - a) = f'(a) \cdot 0 = 0.$$

Consequently, we find that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (f(x) - f(a) + f(a)) = \lim_{x \rightarrow a} (f(x) - f(a)) + \lim_{x \rightarrow a} f(a) = f(a).$$

Hence, by definition, f is continuous at $x = a$. □

Rules of Differentiation

Theorem 2.3 (Derivative of a Constant Function). If c is a constant, then

$$\frac{d}{dx}(c) = 0. \quad (2.13)$$

Proof. Let $f(x) = c$. Then

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0.$$

□

Theorem 2.4 (The Power Rule). If $n \in \mathbb{Z}^+$ and $f(x) = x^n$, then

$$f'(x) = \frac{d}{dx}(x^n) = nx^{n-1}. \quad (2.14)$$

Proof. Let $f(x) = x^n$, where $n \in \mathbb{Z}^+$. Then, we obtain

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{[x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + nxh^{n-1} + h^n] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + nxh^{n-1} + h^n}{h} \\ &= \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \dots + nxh^{n-2} + h^{n-1} \right] \\ &= nx^{n-1}. \end{aligned}$$

□

In general, if $n \in \mathbb{R}$, then $\frac{d}{dx}(x^n) = nx^{n-1}$.

Theorem 2.5 (The Constant Multiple Function). If f is a differentiable function and c is a constant, then

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x). \quad (2.15)$$

Proof. Let $g(x) = cf(x)$. Then,

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= \lim_{h \rightarrow 0} c \left[\frac{f(x+h) - f(x)}{h} \right] \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= cf'(x). \end{aligned}$$

□

Theorem 2.6 (The Sum Rule). If f and g are differentiable functions, then

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x). \quad (2.16)$$

Proof. Let $F(x) = f(x) + g(x)$

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= f'(x) + g'(x). \end{aligned}$$

□

Theorem 2.7 (The Difference Rule). If f and g are differentiable functions, then

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x). \quad (2.17)$$

Proof.

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}[f(x) + (-g(x))] = \frac{d}{dx}f(x) + \frac{d}{dx}[-g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x).$$

[One can also prove this using Definition 2.3.]

□

Theorem 2.8 (The Product Rule). If f and g are differentiable functions, then

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x) \quad (2.18)$$

Proof. Let $F(x) = f(x)g(x)$. Then

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[f(x+h) \left(\frac{g(x+h) - g(x)}{h} \right) + g(x) \left(\frac{f(x+h) - f(x)}{h} \right) \right] \\ &= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} + \lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= f(x)g'(x) + g(x)f'(x). \end{aligned}$$

□

Theorem 2.9 (The Quotient Rule). If f and g are differentiable functions and $g(x) \neq 0$, then

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}. \quad (2.19)$$

Proof. Let $F(x) = \frac{f(x)}{g(x)}$. Then, we get

$$\begin{aligned}
 F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\left(\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} \right)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \frac{g(x)[f(x+h) - f(x)] - f(x)[g(x+h) - g(x)]}{hg(x+h)g(x)} \\
 &= \frac{\lim_{h \rightarrow 0} \left[g(x) \left(\frac{f(x+h) - f(x)}{h} \right) \right] - \lim_{h \rightarrow 0} \left[f(x) \left(\frac{g(x+h) - g(x)}{h} \right) \right]}{\lim_{h \rightarrow 0} g(x+h) \cdot \lim_{h \rightarrow 0} g(x)} \\
 &= \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}
 \end{aligned}$$

□

Let $y = f(x)$. If x changes from a to $(a + \Delta x)$, we defined the increment of y as

$$\Delta y = f(a + \Delta x) - f(a) \quad (2.20)$$

and we have

$$f'(a) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}. \quad (2.21)$$

Denoted by ε the difference between the difference quotient $\frac{\Delta y}{\Delta x}$ and the derivative $f'(a)$, that is,

$$\varepsilon = \frac{\Delta y}{\Delta x} - f'(a). \quad (2.22)$$

It can be seen that

$$\lim_{\Delta x \rightarrow 0} \varepsilon = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} - f'(a) \right) = f'(a) - f'(a) = 0. \quad (2.23)$$

If we define ε to be 0 when $\Delta x = 0$, then ε becomes a continuous function of Δx . Hence, for a differentiable function f , we can write

$$\Delta y = f'(a)\Delta x + \varepsilon\Delta x, \quad (2.24)$$

where $\varepsilon \rightarrow 0$ as $\Delta x \rightarrow 0$ and ε is a continuous function of Δx .

Theorem 2.10 (The Chain Rule). If f and g are both differentiable and $F = f \circ g$ is the composite function defined by $F(x) = f(g(x))$, then F is differentiable and F' is given by

$$F'(x) = f'(g(x))g'(x). \quad (2.25)$$

If we write $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}. \quad (2.26)$$

Proof. Suppose that $u = g(x)$ is differentiable at a and $y = f(u)$ is differentiable at $b = g(a)$. If Δx is an increment in x , and Δu and Δy are the corresponding increments in u and y , then Equation 2.24 gives us

$$\Delta u = g'(a)\Delta x + \varepsilon_1\Delta x, \quad (2.27)$$

where $\varepsilon_1 \rightarrow 0$ as $\Delta x \rightarrow 0$, and

$$\Delta y = f'(b)\Delta u + \varepsilon_2\Delta u, \quad (2.28)$$

where $\varepsilon_2 \rightarrow 0$ as $\Delta u \rightarrow 0$. It follows that,

$$\Delta y = [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1]\Delta x, \quad (2.29)$$

from which we get

$$\frac{\Delta y}{\Delta x} = [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1]. \quad (2.30)$$

We see that

$$\lim_{\Delta x \rightarrow 0} \Delta u = \lim_{\Delta x \rightarrow 0} [g'(a) + \varepsilon_1]\Delta x = 0, \quad (2.31)$$

that is, $\Delta u \rightarrow 0$ as $\Delta x \rightarrow 0$. So, $\varepsilon_1 \rightarrow 0$ and $\varepsilon_2 \rightarrow 0$ as $\Delta x \rightarrow 0$. Hence,

$$\begin{aligned} F'(a) &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1] \\ &= \lim_{\Delta x \rightarrow 0} [f'(b) + \varepsilon_2] \cdot \lim_{\Delta x \rightarrow 0} [g'(a) + \varepsilon_1] \\ &= f'(b)g'(a). \end{aligned}$$

Therefore, $F'(a) = f'(g(a))g'(a)$. □

Theorem 2.11. (General Power Rule) If $n \in \mathbb{R}$ and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}. \quad (2.32)$$

Equivalently,

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x). \quad (2.33)$$

Example 2.4. Find the derivative of the function.

(a) $g(t) = t^3 - 3t + 4$

(b) $h(x) = \left(\frac{x-1}{x+1}\right)^3$

(c) $f(x) = (2x-1)^2(x^2+1)^3$

Solution.



(a) Let $u = g(t) = t^3 - 3t + 4$. Then,

$$\frac{du}{dt} = \frac{d}{dt}(t^3 - 3t + 4) = \frac{d}{dt}(t^3) - 3\frac{d}{dt}(t) + \frac{d}{dt}(4) = 3t^2 - 3 + 0 = 3(t^2 - 1).$$

(b) Let $h(x) = \left(\frac{x-1}{x+1}\right)^3$. Then, we have that

$$h'(x) = 3\left(\frac{x-1}{x+1}\right)^2 \cdot \frac{(1)(x+1) - (1)(x-1)}{(x+1)^2} = \frac{6(x-1)^2}{(x+1)^4}.$$

(c) Let $f(x) = (2x-1)^2(x^2+1)^3$. Then, we get

$$\begin{aligned} f'(x) &= 2(2x-1)(2)(x^2+1)^3 + (2x-1)^2(3)(x^2+1)^2(2x) \\ &= 2(2x-1)(x^2+1)^2(8x^2-3x+2). \end{aligned}$$

■

Derivatives of Trigonometric Functions

Theorem 2.12 (Derivative of $\sin(x)$).

$$\frac{d}{dx}(\sin(x)) = \cos(x). \quad (2.34)$$

Proof. Let $f(x) = \sin(x)$. Then,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{\sin(x)\cos(h) - \sin(x)}{h} + \frac{\sin(h)\cos(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\sin(x) \left(\frac{\cos(h) - 1}{h} \right) + \cos(x) \left(\frac{\sin(h)}{h} \right) \right] \\ &= (\sin(x))(0) + (\cos(x))(1) \\ &= \cos(x). \end{aligned}$$

□

Theorem 2.13 (Derivative of $\cos(x)$).

$$\frac{d}{dx}(\cos(x)) = -\sin(x). \quad (2.35)$$

Proof. Let $f(x) = \cos(x)$. Then,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h}. \end{aligned}$$

This leads us to

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \left[\frac{\cos(x)\cos(h) - \cos(x)}{h} - \frac{\sin(x)\sin(h)}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\cos(x) \left(\frac{\cos(h) - 1}{h} \right) - \sin(x) \left(\frac{\sin(h)}{h} \right) \right] \\ &= (\cos(x))(0) - (\sin(x))(1) \\ &= -\sin(x). \end{aligned}$$

□

Example 2.5. Find

$$\frac{d}{dx}(\tan(x)).$$

Solution. We have that

$$\frac{d}{dx}(\tan(x)) = \frac{d}{dx} \left(\frac{\sin(x)}{\cos(x)} \right) = \frac{\cos(x)\cos(x) - (-\sin(x))\sin(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x).$$

■

Derivatives of the other trigonometric functions are

$$\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x), \quad \frac{d}{dx}(\sec(x)) = \sec(x)\tan(x), \quad \frac{d}{dx}(\cot(x)) = -\csc^2(x).$$

Example 2.6. Find y' , given that

- (a) $y = \cot^2(\sin(\theta))$
- (b) $y = \sin(\tan(\sqrt{\sin(x)}))$

Solution.

- (a) We have that

$$\begin{aligned} y' &= \frac{d}{d\theta} (\cot^2(\sin(\theta))) = 2\cot(\sin(\theta)) \frac{d}{d\theta} [\cot(\sin(\theta))] \\ &= 2\cot(\sin(\theta)) \left[-\csc^2(\sin(\theta)) \frac{d}{d\theta} (\sin(\theta)) \right] \\ &= 2\cot(\sin(\theta)) [-\csc^2(\sin(\theta)) \cos(\theta)] \\ &= -2\cot(\sin(\theta)) \csc^2(\sin(\theta)) \cos(\theta). \end{aligned}$$

(b) Let $y = \sin(\tan(\sqrt{\sin(x)}))$. We obtain

$$\begin{aligned}\frac{dy}{dx} &= \cos(\tan(\sqrt{\sin(x)})) \frac{d}{dx}(\tan(\sqrt{\sin(x)})) \\ &= \cos(\tan(\sqrt{\sin(x)})) \sec^2(\sqrt{\sin(x)}) \frac{d}{dx}(\sqrt{\sin(x)}) \\ &= \cos(\tan(\sqrt{\sin(x)})) \sec^2(\sqrt{\sin(x)}) \cdot \frac{1}{2}(\sin(x))^{-\frac{1}{2}} \cos(x).\end{aligned}$$

■

Example 2.7. Find the derivatives of the functions

(a) $y = x^2 \sin^2(2x^2)$

(b) $r = \left(\frac{\sin \theta}{\cos \theta - 1}\right)^2$

Solution.

(a) For $y = x^2 \sin^2(2x^2)$, we obtain

$$\begin{aligned}\frac{dy}{dx} &= 2x \sin^2(2x^2) + x^2 \cdot 2 \sin(2x^2) \cos(2x^2) \cdot 4x \\ &= 2x \sin^2(2x^2) + 8x^3 \sin(2x^2) \cos(2x^2).\end{aligned}$$

(b) For $r = \left(\frac{\sin(\theta)}{\cos(\theta) - 1}\right)^2$, we get

$$\begin{aligned}\frac{dr}{d\theta} &= 2 \left(\frac{\sin(\theta)}{\cos(\theta) - 1}\right) \cdot \frac{(\cos(\theta) - 1) \cos(\theta) - \sin(\theta)(-\sin(\theta))}{(\cos(\theta) - 1)^2} \\ &= \frac{2 \sin(\theta)}{\cos(\theta) - 1} \cdot \frac{\cos^2(\theta) - \cos(\theta) + \sin^2(\theta)}{(\cos(\theta) - 1)^2} \\ &= \frac{2 \sin(\theta)(1 - \cos(\theta))}{(\cos(\theta) - 1)^3} \\ &= \frac{-2 \sin(\theta)}{(\cos(\theta) - 1)^2}.\end{aligned}$$

■

Implicit Differentiation

Suppose that a function is defined implicitly via an equation in x and y . To compute $\frac{dy}{dx}$

1. Differentiate both sides of the equation with respect to x . Make sure that the derivative of any term involving y includes the factor $\frac{dy}{dx}$.
2. Solve the resulting equation for $\frac{dy}{dx}$ in terms of x and y

Example 2.8. Find $\frac{dy}{dx}$ if $y^4 - 2y^3 + x^3y^2 - \cos(x) = 8$

Solution. Differentiating implicitly, we find that

$$4y^3 \frac{dy}{dx} - 6y^2 \frac{dy}{dx} + 3x^2y^2 + x^3 \cdot 2y \frac{dy}{dx} + \sin(x) = 0,$$

which produces

$$\frac{dy}{dx} = \frac{-3x^2y^2 - \sin(x)}{4y^3 - 6y^2 + 2yx^3}.$$

■

Example 2.9. Find $\frac{dy}{dx}$ if $\sqrt{xy} = x^2 + 2y^2$

Solution. We get

$$\frac{1}{2}(xy)^{-\frac{1}{2}} \left(y + x \frac{dy}{dx} \right) = 2x + 4y \frac{dy}{dx},$$

which yields

$$\frac{dy}{dx} = \left(2x - \frac{y}{2\sqrt{xy}} \right) \left(\frac{2\sqrt{xy}}{x - 8y\sqrt{xy}} \right) = \frac{4x\sqrt{xy} - y}{x - 8y\sqrt{xy}}.$$

■

Derivatives of Exponential Functions

Let $f(x) = a^x$, where $a \in \mathbb{R}^+ \setminus \{1\}$. We find that

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h \rightarrow 0} a^x \left(\frac{a^h - 1}{h} \right) \\ &= a^x \lim_{h \rightarrow 0} \frac{a^{0+h} - a^0}{h} \\ &= a^x f'(0). \end{aligned}$$

Definition 2.6 (The Number e). e is the real number such that

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1. \quad (2.36)$$

From Definition 2.6, we see that if $f(x) = e^x$, then we have that

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1. \quad (2.37)$$

Theorem 2.14 (Derivative of e^x).

$$\frac{d}{dx}(e^x) = e^x \quad (2.38)$$

More generally, let $y = e^{g(x)}$. Then, we get

$$\frac{dy}{dx} = e^{g(x)} g'(x), \quad (2.39)$$

where we have used the chain rule.

Example 2.10. Find the derivative of the given function.

(a) $y = xe^{-x^2}$

(b) $y = e^{x \cos(x)}$

(c) $y = \frac{e^{2u}}{e^u + e^{-u}}$

Solution.

(a) Let $y = xe^{-x^2}$. Then, we find that

$$\frac{dy}{dx} = e^{-x^2} + xe^{-x^2}(-2x) = e^{-x^2}(1 - 2x^2).$$

(b) For $y = e^{x \cos x}$, we obtain

$$\frac{dy}{dx} = e^{x \cos(x)}(\cos(x) - x \sin(x)).$$

(c) Let $y = \frac{e^{2u}}{e^u + e^{-u}}$. We deduce that

$$\begin{aligned} \frac{dy}{du} &= \frac{2e^{2u}(e^u + e^{-u}) - e^{2u}(e^u - e^{-u})}{(e^u + e^{-u})^2} \\ &= \frac{e^{2u}(2e^u + 2e^{-u} - e^u + e^{-u})}{(e^u + e^{-u})^2} \\ &= \frac{e^{2u}(e^u + 3e^{-u})}{(e^u + e^{-u})^2}. \end{aligned}$$

■

Now, let $a \in \mathbb{R}^+ \setminus \{1\}$. We observe that $a^x = (e^{\ln(a)})^x = e^{x \ln(a)}$. Consequently, we obtain

$$\frac{d}{dx}(a^x) = \frac{d}{dx}(e^{x \ln(a)}) = e^{x \ln(a)} \frac{d}{dx}(x \ln(a)) = e^{x \ln(a)} \cdot \ln(a) = a^x \ln(a).$$

Therefore, we infer

$$\frac{d}{dx}(a^x) = a^x \ln(a). \quad (2.40)$$

If $y = a^{g(x)}$, then we have that

$$\frac{d}{dx}(a^{g(x)}) = a^{g(x)}(\ln(a))g'(x). \quad (2.41)$$

Example 2.11. Find the derivative of the function



(a) $y = 10^{1-x^2}$

(b) $y = 2^{3x^2}$

(c) $y = 2^{\sin(\pi x)}$

Solution.

(a) Let $y = 10^{1-x^2}$. Then, it follows that

$$\frac{dy}{dx} = 10^{1-x^2} \ln(10)(-2x) = -2x \ln(10) 10^{1-x^2}.$$

(b) Let $y = 2^{3x^2}$. Then, we obtain

$$\begin{aligned} \frac{dy}{dx} &= 2^{3x^2} \ln(2) \frac{d}{dx}(3x^2) = 2^{3x^2} \ln(2) \left[3x^2 \ln(3) \frac{d}{dx}(x^2) \right] \\ &= 2^{3x^2} \ln(2) \left[3x^2 \ln(3)(2x) \right] \\ &= 2x \ln(2) \ln(3) 2^{3x^2} 3x^2 \end{aligned}$$

(c) Let $y = 2^{\sin(\pi x)}$. Then, we get

$$\begin{aligned} \frac{dy}{dx} &= 2^{\sin(\pi x)} \ln(2) \frac{d}{dx}(\sin(\pi x)) = \ln(2) 2^{\sin(\pi x)} \cos(\pi x) \frac{d}{dx}(\pi x) \\ &= \pi \ln(2) 2^{\sin(\pi x)} \cos(\pi x). \end{aligned}$$

■

Derivatives of Logarithmic Functions

Theorem 2.15 (Derivative of $\ln(x)$).

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x} \quad (2.42)$$

Proof. Let $y = \ln(x)$. Then, we see that $e^y = x$, from which we get $e^y \frac{dy}{dx} = 1$. It follows that

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x},$$

since $e^y = x$. □

More generally, let $y = \ln[g(x)]$. Then, we get $e^y = g(x)$. As a result, we get $e^y \frac{dy}{dx} = g'(x)$, which yields

$$\frac{dy}{dx} = \frac{g'(x)}{e^y} = \frac{g'(x)}{g(x)}. \quad (2.43)$$

Example 2.12. Find $f'(x)$ if $f(x) = \ln(|x|)$.

Solution. We have that

$$f(x) = \begin{cases} \ln(x), & \text{if } x > 0 \\ \ln(-x), & \text{if } x < 0 \end{cases}.$$

This leads us to

$$f'(x) = \begin{cases} \frac{1}{x}, & \text{if } x > 0 \\ \frac{-1}{-x} = \frac{1}{x}, & \text{if } x < 0 \end{cases}.$$

Therefore, $f'(x) = \frac{1}{x}$ for all $x \neq 0$. ■

Example 2.13. Differentiate f given that

(a) $f(x) = x^2 \ln(1 - x^2)$

(b) $f(x) = \ln(x^4 \sin^2 x)$

Solution.

(a) Let $f(x) = x^2 \ln(1 - x^2)$. We deduce that

$$f'(x) = 2x \ln(1 - x^2) + x^2 \left(\frac{-2x}{1 - x^2} \right) = 2x \ln(1 - x^2) - \frac{2x^3}{1 - x^2}.$$

(b) We obtain

$$f'(x) = \frac{4x^3 \sin^2(x) + (x^4)2 \sin(x) \cos(x)}{x^4 \sin(x)} = \frac{4 \sin(x) + 2x \cos(x)}{x \sin(x)}.$$
■

Now, let us consider $y = \log_a(x)$, where $a > 0$ and $a \neq 1$. Then, we see that $a^y = x$. Differentiating implicitly, we arrive at $a^y \ln(a) \frac{dy}{dx} = 1$, which produces

$$\frac{dy}{dx} = \frac{1}{a^y \ln(a)} = \frac{1}{x \ln(a)}.$$

That is,

$$\frac{d}{dx}(\log_a(x)) = \frac{1}{x \ln(a)}. \quad (2.44)$$

A similar argument yields

$$\frac{d}{dx}(\log_a g(x)) = \frac{g'(x)}{g(x) \ln(a)}. \quad (2.45)$$

Example 2.14. Differentiate

(a) $f(x) = \log_{10}(2 + \sin(x))$

(b) $g(x) = \log_{10} \left(\frac{x}{x-1} \right)$

Solution.

(a) We find that

$$f'(x) = \frac{\cos(x)}{(2 + \sin(x)) \ln(10)}.$$

(b) We obtain

$$g'(x) = \frac{\left[\frac{(1)(x-1) - x(1)}{(x-1)^2} \right]}{\left[\frac{x \ln(10)}{x-1} \right]} = \left(\frac{-1}{(x-1)^2} \right) \left(\frac{x-1}{x \ln(10)} \right) = \frac{-1}{x(x-1) \ln(10)}.$$

■

Derivative of a function in Parametric form

Definition 2.7 (Parametric Curve). If x and y are given as functions

$$x = f(t), \quad y = g(t)$$

over an interval of t -values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a parametric curve. The equations are parametric equations of the curve.

Now, if f and g are differentiable at t and y is a differentiable function of x where $\frac{dx}{dt} \neq 0$, then

$$\frac{dy}{dx} = \left(\frac{dy}{dt} \right) / \frac{dx}{dt}, \quad (2.46)$$

which can be expressed as

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}. \quad (2.47)$$

Example 2.15. Find $\frac{dy}{dx}$ given that

$$x = t - t^2 \quad \text{and} \quad y = t - t^3.$$

Solution. We obtain

$$\frac{dx}{dt} = 1 - 2t \quad \text{and} \quad \frac{dy}{dt} = 1 - 3t^2,$$

which imply that

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1 - 3t^2}{1 - 2t}.$$

■

Higher Derivatives

The second derivative of a function is the derivative of the derivative of f , that is, $(f')' = f''$. Let $y = f(x)$. Then

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2} = f''(x) = D^2f(x) = y''.$$

The third derivative f''' is the derivative of the second derivative. Let $y = f(x)$. Then

$$y''' = f'''(x) = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d^3y}{dx^3} = D^3f(x).$$

In general, the n^{th} derivative of f is denoted by $f^{(n)}$. If $y = f(x)$, then

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n} = D^n f(x).$$

Example 2.16. Find $\frac{d^2y}{dx^2}$ given that $x = t - \sin(t)$ and $y = 1 - \cos(t)$

Solution. We find that

$$\frac{dx}{dt} = 1 - \cos(t) \quad \text{and} \quad \frac{dy}{dt} = \sin(t),$$

from which we get

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} / \frac{dx}{dt} = \frac{\sin(t)}{1 - \cos(t)}.$$

Furthermore, we obtain

$$\begin{aligned} \frac{d}{dt} \left(\frac{dy}{dx} \right) &= \frac{\cos(t)(1 - \cos(t)) - \sin(t)\sin(t)}{(1 - \cos(t))^2} \\ &= \frac{\cos(t) - \cos^2 t - \sin^2(t)}{(1 - \cos(t))^2} \\ &= \frac{\cos(t) - 1}{(1 - \cos(t))^2}. \end{aligned}$$

We have that

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx} = \frac{d}{dt} \left(\frac{dy}{dx} \right) / \frac{dx}{dt}.$$

It follows that

$$\frac{d^2y}{dx^2} = \frac{\cos(t) - 1}{(1 - \cos(t))^2} \times \frac{1}{1 - \cos(t)} = \frac{-1}{(1 - \cos(t))^2}.$$

■

Example 2.17. If $f(x) = e^x - x^3$, find $f'(x)$ and $f''(x)$.

Solution. We get

$$f'(x) = e^x - 3x^2,$$

which leads us to

$$f''(x) = e^x - 6x.$$

■

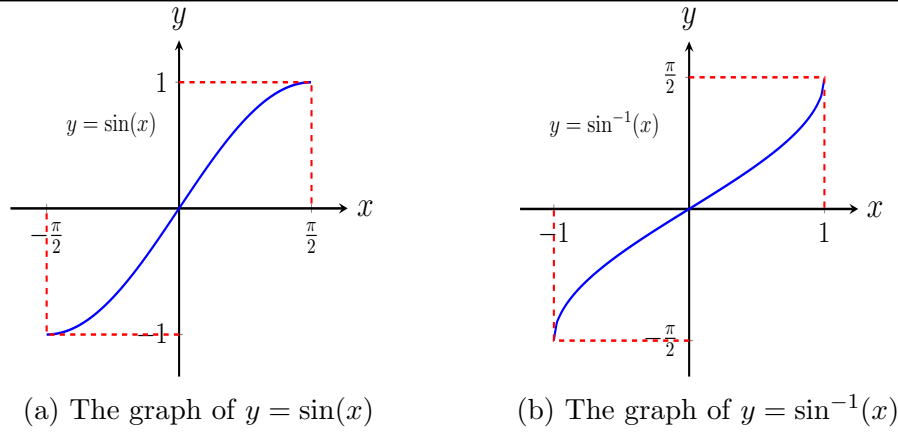


Figure 2.3

Derivatives of Inverse Trigonometric Functions

Let $y = \sin^{-1}(x)$. Then, $\sin(y) = x$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. Differentiating implicitly, we obtain $\cos(y) \frac{dy}{dx} = 1$, which means that $\frac{dy}{dx} = \frac{1}{\cos(y)}$. We note that $\cos(y) \geq 0$, since $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. This leads us to $\cos(y) = \sqrt{1 - \sin^2(y)} = \sqrt{1 - x^2}$. It follows that $\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}$. Therefore, we have that

$$\frac{d}{dx}(\sin^{-1}(x)) = \frac{1}{\sqrt{1 - x^2}}. \quad (2.48)$$

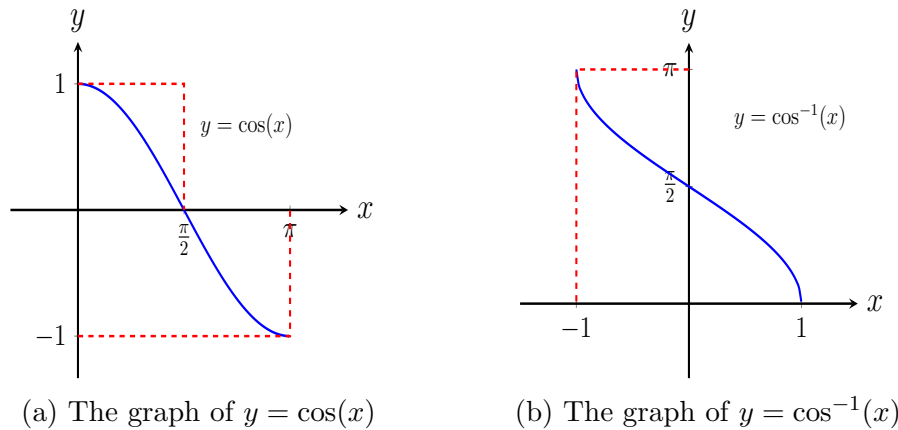


Figure 2.4

Let $y = \cos^{-1}(x)$. Then, we have that $\cos(y) = x$ and $0 \leq y \leq \pi$. We see that $-\sin(y) \frac{dy}{dx} = 1$, and so $\frac{dy}{dx} = \frac{-1}{\sin(y)}$. Besides, $\sin(y) \geq 0$ since $0 \leq y \leq \pi$. Thus,

$\sin(y) = \sqrt{1 - \cos^2(y)} = \sqrt{1 - x^2}$. Therefore, we get

$$\frac{d}{dx}(\cos^{-1}(x)) = \frac{-1}{\sqrt{1 - x^2}}. \quad (2.49)$$

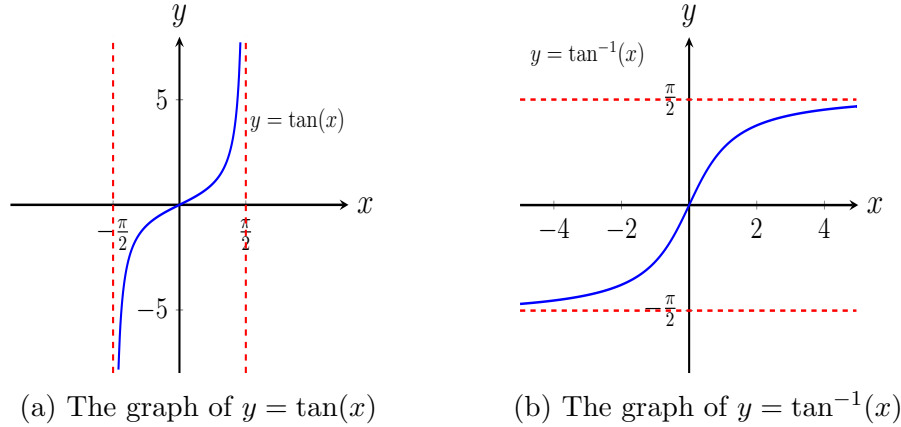


Figure 2.5

Let $y = \tan^{-1}(x)$. Then, we have that $\tan(y) = x$ where $-\frac{\pi}{2} < y < \frac{\pi}{2}$. It follows that $\sec^2(y) \frac{dy}{dx} = 1$, and so

$$\frac{dy}{dx} = \frac{1}{\sec^2(y)} = \frac{1}{1 + \tan^2(y)} = \frac{1}{1 + x^2}.$$

Therefore,

$$\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1 + x^2}. \quad (2.50)$$

The others are:

$$\frac{d}{dx}(\csc^{-1}(x)) = \frac{-1}{x\sqrt{x^2 - 1}}, \quad \frac{d}{dx}(\sec^{-1}(x)) = \frac{1}{x\sqrt{x^2 - 1}}, \quad \frac{d}{dx}(\cot^{-1}(x)) = \frac{-1}{1 + x^2}.$$

Example 2.18. Find the derivatives of

- (a) $f(x) = \cos^{-1}(3x)$
- (b) $g(x) = \tan^{-1}(\sqrt{2x + 3})$
- (c) $y = \sec^{-1}(e^{-2x})$

Solution.

- (a) Let $f(x) = \cos^{-1}(3x)$. Then, we obtain

$$f'(x) = \frac{-1}{\sqrt{1 - (3x)^2}} \times 3 = \frac{-3}{\sqrt{1 - 9x^2}}.$$

(b) Let $g(x) = \tan^{-1}(\sqrt{2x+1})$. Then, we deduce that

$$\begin{aligned} g'(x) &= \frac{1}{1 + (\sqrt{2x+1})^2} \times \frac{1}{2}(2x+1)^{-\frac{1}{2}} \times 2 \\ &= \frac{1}{4+2x} \times \frac{1}{\sqrt{2x+1}} \\ &= \frac{1}{2(x+2)\sqrt{2x+1}}. \end{aligned}$$

(c) Let $y = \sec^{-1}(e^{-2x})$. Then, we get

$$\frac{dy}{dx} = \frac{1}{(e^{-2x})\sqrt{(e^{-2x})^2 - 1}} \cdot -2e^{-2x} = \frac{-2}{\sqrt{e^{-4x} - 1}}.$$

■

Logarithmic Differentiation

1. Take the natural logarithm of both sides of an equation $y = f(x)$ and use the laws of logarithm to simplify it.
2. Differentiate implicitly with respect to x
3. Solve the resulting equation for y' .

If $f(x) < 0$ for some values of x , then $\ln(f(x))$ is undefined. However, we can write $|y| = |f(x)|$ and use the fact that

$$\frac{d}{dx}(\ln |g(x)|) = \frac{g'(x)}{g(x)},$$

where $g(x) \neq 0$ for all x .

Example 2.19. If n is any real number and $f(x) = x^n$, show that $f'(x) = nx^{n-1}$.

Solution. Let $y = x^n$. Then, $|y| = |x^n| = |x|^n$, which implies that

$$\ln(|y|) = \ln(|x|^n) = n \ln(|x|), \quad x \neq 0.$$

This leads to $\frac{y'}{y} = \frac{n}{x}$, from which we get

$$y' = n \frac{y}{x} = \frac{nx^n}{x} = nx^{n-1}.$$

■

Example 2.20. Differentiate $y = x^{\sqrt{x}}$.

Solution. From $y = x^{\sqrt{x}}$, we see that $\ln(y) = \sqrt{x} \ln(x)$. Hence, we obtain

$$\frac{y'}{y} = \frac{1}{2}x^{-\frac{1}{2}} \ln(x) + \sqrt{x} \cdot \frac{1}{x} = \frac{\ln(x)}{2\sqrt{x}} + \frac{\sqrt{x}}{x},$$

which produces

$$y' = \left(\frac{\ln(x)}{2\sqrt{x}} + \frac{1}{\sqrt{x}} \right) x^{\sqrt{x}} = \left(\frac{\ln(x) + 2}{2\sqrt{x}} \right) x^{\sqrt{x}}.$$

■

Example 2.21. Use logarithmic differentiation to find the derivative of the function

$$y = \sqrt[4]{\frac{x^2 + 1}{x^2 - 1}}.$$

Solution. Taking logarithm of both sides of the given equation, we get

$$\ln y = \frac{1}{4} [\ln(x^2 + 1) - \ln(x^2 - 1)],$$

which yields

$$\frac{y'}{y} = \frac{1}{4} \left[\frac{2x}{x^2 + 1} - \frac{2x}{x^2 - 1} \right] = \frac{1}{4} \left[\frac{2x^3 - 2x - 2x^3 - 2x}{x^4 - 1} \right] = \frac{1}{4} \left[\frac{-4x}{x^4 - 1} \right].$$

Therefore, we have that

$$y' = \frac{-x}{x^4 - 1} \sqrt[4]{\frac{x^2 + 1}{x^2 - 1}}.$$

■

EXERCISE TWO

- Find from the first principles the derivatives of the following functions

(a) $f(x) = \frac{1}{(x-1)^2}$

(b) $f(x) = \sqrt{x+1}$

(c) $f(x) = \frac{4-3x}{2+x}$

(d) $f(x) = \sin(3x)$

- Differentiate the following functions with respect to x .

(a) $f(x) = \frac{x^2 + 4x + 3}{\sqrt{x}}$

(b) $f(x) = \frac{\sqrt{x} - 1}{\sqrt{x} + 1}$

(c) $f(x) = (6x^2 + 5)^3(x^3 - 7)^4$

(d) $f(x) = \sqrt{x^2 + 1}$

(e) $f(x) = (5 - 2x^2)^{-1}$

- Find the derivative of each of the following functions:

(a) $f(x) = \tan(3x) - 4\sec(5x)$

(b) $f(x) = \frac{\sin^2(x)}{\cos(x)}$

(c) $f(x) = \sqrt{1 + 2\tan(x)}$

(d) $f(x) = \sin^3(x) + \cos^3(x)$

(e) $f(x) = \sec^2(2x) - \tan^2(2x)$

(f) $f(x) = \frac{\sin(x) + \cos(x)}{\sin(x) - \cos(x)}$

4. If $F(x) = f(g(x))$, $g(3) = 6$, $g'(3) = 4$ and $f'(6) = 7$, find $F'(3)$.

5. (a) If $y = \sec(x) \tan(x)$, show that

$$\frac{dy}{dx} = 2 \sec^3(x) - \sec(x).$$

(b) If $y = x \sin(x)$, show that

$$\frac{d^2y}{dx^2} = 2 \cos(x) - y.$$

6. (a) For what values of A and B does $y = A \cos(x) + B \sin(x)$ satisfy the equation

$$y'' + 2y' + 3y = 2 \sin(x).$$

(b) For what values of A and B does $y = Ax \cos(x) + Bx \sin(x)$ satisfy the equation

$$y'' + y = -3 \cos(x).$$

7. If the equation $2x^2 + 6xy + 4y^2 = 3$ defines y as a twice-differentiable function of x , express $\frac{dy}{dx}$ in terms of x and y and show that

$$\frac{d^2y}{dx^2} = \frac{3}{(3x + 4y)^3}.$$

8. If the equation $y^4 = 4xy + 3$ defines y as a twice-differentiable function of x , express $\frac{dy}{dx}$ in terms of x and y and show that

$$\frac{d^2y}{dx^2} = \frac{3(1 - y^4)}{2(y^3 - x)^3}.$$

9. Find $\frac{dy}{dx}$ in terms of t when

$$x(t) = \frac{1+t}{1-2t}, \quad y(t) = \frac{1+2t}{1-t}.$$

Prove that $\frac{dy}{dx} = 1$ when $t = 0$ and find a second value of t for which $\frac{dy}{dx} = 1$. Prove that

$$\frac{d^2y}{dx^2} = -\frac{2}{3} \left(\frac{1-2t}{1-t} \right)^3$$

10. For the curve $x = t^2 + t$, $y = 3 - t^3$, show that

$$\frac{d^2y}{dx^2} = -\frac{6t(t+1)}{(2t+1)^3}, \quad t \neq -1/2.$$

11. Differentiate the following with respect to x .

- (a) $f(x) = \sin^{-1}(x/3)$
- (b) $f(x) = \cos^{-1}(x)$
- (c) $f(x) = \tan^{-1}(\cot(x))$
- (d) $f(x) = (1 + x^2) \tan^{-1}(x)$
- (e) $f(x) = x \cos^{-1}(x)$
- (f) $f(x) = \sqrt{1 - x^2} \sin^{-1}(x)$

12. (a) If $y = (\sin^{-1}(x))^2$, prove that

$$(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2.$$

(b) Given that $y = \sin(k \sin^{-1}(x))$, where k is a constant, prove that

$$(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + k^2 y = 0.$$

13. Differentiate the following functions with respect to x .

- (a) $f(x) = e^{2x}$
- (b) $f(x) = e^{2x^3}$
- (c) $f(x) = e^{x \cos(x)}$
- (d) $f(x) = e^{2x} \cos(3x)$
- (e) $f(x) = x^2 e^{3x-2}$
- (f) $f(x) = e^{x^2} \sec(x)$.

14. Differentiate the following functions with respect to x .

- (a) $f(x) = \ln(x^3 - 3x)$
- (b) $f(x) = \ln(\sec(x) + \tan(x))$
- (c) $f(x) = \ln(\sqrt{4x + 5})$
- (d) $f(x) = \ln\left(\frac{\cos(x)}{\sqrt{1 - x^2}}\right)$
- (e) $f(x) = \ln(3x^2 e^{-x})$

15. If $x = 1 + \ln(t)$ and $y = \ln(t + 1)$, find $\frac{dy}{dx}$ in terms of t .

16. Differentiate the following functions with respect to x .

- (a) $f(x) = \log_{10} x$
- (b) $f(x) = 5^{2x}$
- (c) $f(x) = x^{\ln(x)}$
- (d) $f(x) = \sqrt[3]{\frac{x^2 - 2}{x^2 + 4}}$